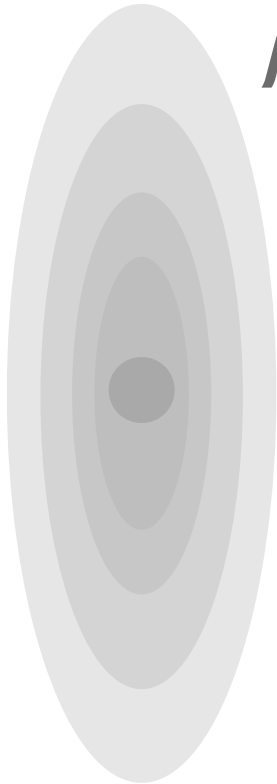


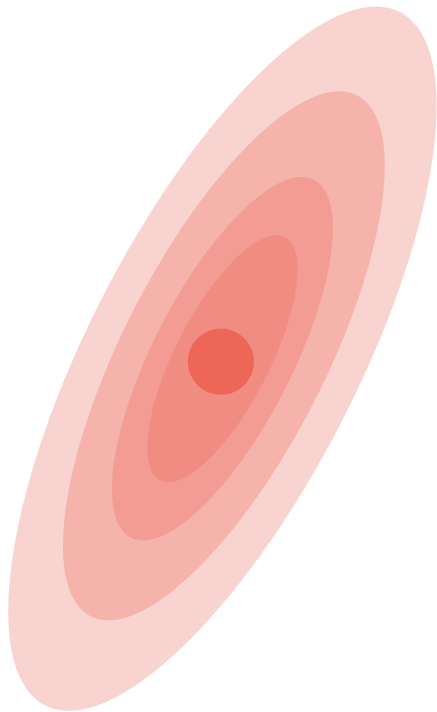
α



ρ_t



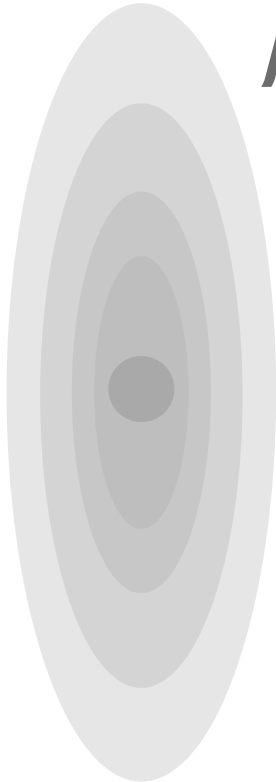
β



α



ρ_t



β

$$W_2^2(\alpha, \beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \rho_t(x) \, dx \, dt$$

where ρ_t, v_t : measure, vector field. Subject to:

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0 \quad \text{(continuity equation)}$$

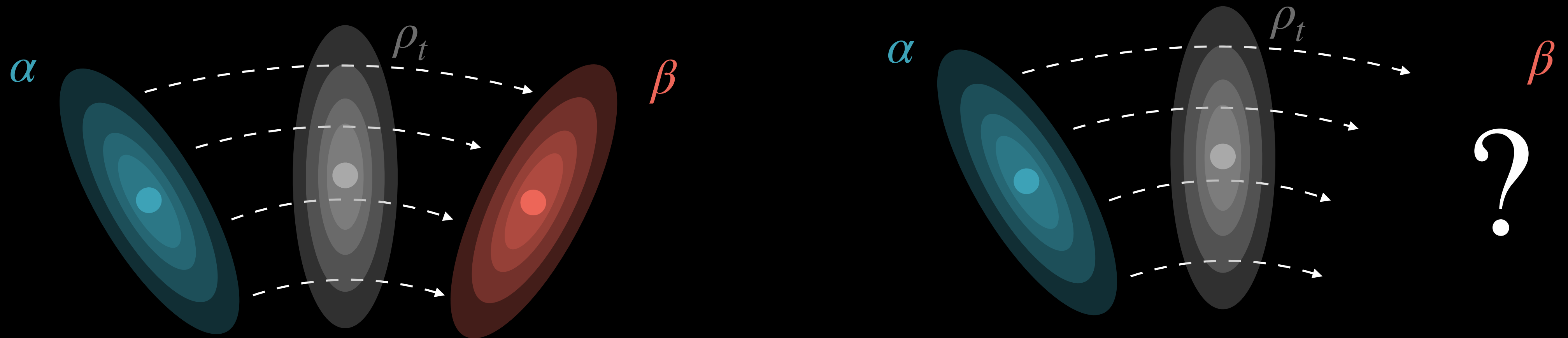
$$\rho_0 = \alpha, \rho_1 = \beta \quad \text{(boundary conditions)}$$

Why gradient flow?

- Benamou-Brenier formulation:
- What if finding the entire path $\{\rho_t, v_t\}$ is too costly?
- What if we don't *explicitly* know β ?

$$W_2^2(\alpha, \beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \|v_t(x)\|^2 \rho_t(x) dx dt$$

where ρ_t, v_t : measure, vector field. Subject to:
 $\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$ (continuity equation)
 $\rho_0 = \alpha, \rho_1 = \beta$ (boundary conditions)



Gradient Flows Intuition